

## Document Q&A RAG System - Explanation Document

### 1. Design Parameter:

Chunk Size (800 characters) and Overlap (150 characters)

I used fixed-size character chunking rather than token- or sentence-based chunking, and settled on 800 characters with 150 characters of overlap

Why this value: Chunk size is a direct trade-off between two failure modes. Too small, and a chunk loses the surrounding context needed for the embedding to represent what it's actually about, a sentence fragment about "the policy" without the paragraph naming which policy embeds ambiguously and retrieves poorly. Too large, and the chunk mixes multiple topics into one embedding, diluting the similarity signal for any single question, and wastes context window on irrelevant text once retrieved. 800 characters (roughly 120 - 150 words) is large enough to hold a complete thought or two but small enough to stay topically coherent for most prose documents.

The 150-character overlap exists specifically to handle facts that fall near a chunk boundary. Without overlap, a sentence split exactly at the cut point can become unretrievable from either half. I chose overlap as roughly 19% of chunk size as a middle ground enough to catch boundary-split content without meaningfully increasing storage/embedding cost.

### 2. Failure Observed:

I had provided a pdf with some old AI/ML notes of mine and I asked it one question: What does prompting mean to an LLM ? with top\_k chunks as 4. It gave a reply saying "I can't find an answer to this in the uploaded documents." And then when I asked it a question: What does fine - tuning mean? It gave the response : "Fine-Tuning is taking a pre-trained model and training it further on a specific dataset (like medical records or legal documents) so it becomes an expert in that domain."

So I think the failure I observed is due to the larger amount of chunks and not being able to retrieve the correct embeddings properly.

### 3. Metrics Measured:

I asked 5 questions whose answers are there in the doc and the 2 questions whose answer is not there in the doc and verified the answers i got:

5/5 answerable questions were correctly grounded and accurate;

2/2 unanswerable questions were correctly refused.

### 4. What I Didn't Finish, and Next Steps

Not finished:

No OCR fallback - scanned/image-only PDFs return no extractable text and are rejected rather than handled.

No formal retrieval-accuracy evaluation harness - correctness was spot-checked manually rather than measured against a labeled question set.

No de-duplication - re-uploading the same file indexes it again as a second, redundant set of chunks.

Chunking is character-based, not sentence- or token-aware, so a chunk can occasionally split mid-sentence.

Next steps, in priority order:(1) build a small labeled set of 15--20 question/expected-answer pairs from a test document and compute retrieval precision automatically, since manual spot-checking doesn't scale and was the weakest part of my validation process; (2) add a de-duplication check on upload (hash the file content, skip or replace on re-upload); (3) move to sentence-aware chunking using a lightweight sentence splitter to reduce mid-sentence cuts while keeping the same size/overlap logic.